1. Risk Strategy Recommendations

1.1. Introduction

The preceding decision trees analysis displays transition from a homogenous MSME dataset into four distinct business risk profiles. The Machine Learning Modelling done in Part 1 of this assignment demonstrates factors such as financial distress, business tenure and business geographic location don't just act in isolation. They compound to define a business operational reality and its associated credit risk. The objective of this section is to translate the analysis done into actionable credit policies. Since there exists diverse risk profiles, deploying a universal lending strategy would be ineffective. The following recommendations propose tailored mitigation tactics designed to protect the lender while appropriately serving each segment. We shall first review the segments in a generic way then deep dive per country.

1.2. Established Veterans Segment

This segment represents the most mature and stable demographic within the MSME portfolio. Based on the analysis, the foundational requirement for this segment is business survival. The dominant and most predictable businesses in the segment consist of highly mature businesses, that have over 19 years of operation and are managed by older people, who maintain manageable financial stress scores that are less than 4.00. The model also flags another demographic, older businesses above 24 years of operation, operated by younger managers who are below 48 years. This implies inherited businesses or second-generation handover. On the other hand, when mature businesses display a high financial distress of above 4.00, they remain in this category only if the owner is older, pointing toward a business that is aging out alongside its manager.

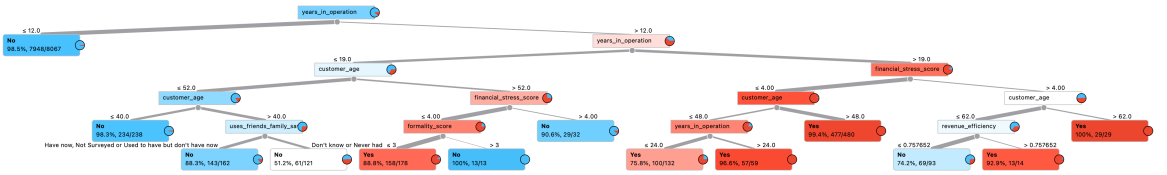

Figure 1: Binary Decision Tree for Established Veterans

The primary risk associated with this segment is stagnation and succession risk. Since these businesses have weathered multiple economic cycles, we pivot from traditional default mitigation to long-term relationship management and focus on continuity. We propose the below:

1.Succession-Linked Financing: For any long-term loan facility, extending beyond a 36-month tenure, we need to mandate a documented designated secondary operator to be listed on the loan account. That is, a younger or junior partner to mitigate the risk identified in older-operator demographic.

2. Digital Transition Support: To counter technological stagnation, especially for the older owners above 60 years, we suggest providing dedicated relationship managers to assist in transitioning these businesses into modern digital payment platforms, thereby protecting the business's operations against younger, tech-savvy competitors. This ensures their businesses stay operational over time.

1.3. High Risk Informal Segment

This segment represents the most vulnerable demographic in the portfolio. The defining feature of this risk category is credit visibility, defined by the active loan holder status. As guided in the notebook, businesses classified as not surveyed mean that they weren't included in the survey, for a particular variable. The dominant profile for this segment is defined by a combination of data invisibility and lack of proven tenure. Businesses that have not been surveyed and have been operating for 19 years or less prove to carry a massive 96.4% probability of being high risk. Moreover, another profile emerges among older, moderately mature businesses. These businesses fall into the high risk category if the owner is below 62 years of age and if the business is displaying significant financial distress determined by the financial stress score.

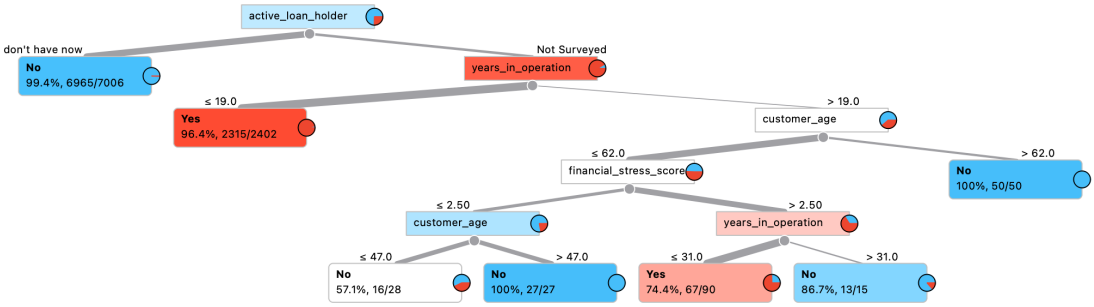

Figure 2: Binary Decision Tree for High Risk Informal

The primary risk associated with this segment is extreme probability to default driven by information asymmetry and unproven business survival chances. We propose the below:

1.Tranche Disbursement Financing: For the younger and untracked businesses that are dominant in this segment, credit is to be extended using a strict micro-stepping model whereby the initial loans are kept minimal. The businesses in this segment will only unlock higher limits after they demonstrate a consecutive 3-5 cycle repayment history.

2.Alternative Data Underwriting: Since there exists a lot of businesses that are tagged under the not surveyed criteria by formal credit systems, the lender should substitute the traditional credit scores with alternative sources of cash-flow data. This can rely on mobile money, utility payment consistency, transaction histories or supplier remittance records to determine the businesses liquidity.

1.4. Large Informal Segment

This segment represents the businesses that have achieved longevity without necessarily adopting formal financial structures. The key determinant of this segment is digital financial literacy and access. The root node splits entirely on the has_internet_banking variable. The dominant profile within the segment, capturing a massive 98.5% probability is defined by lack of operational formality. Specifically, these are businesses operating for 16 years or less, possess a high informal structure formality_score and run by operators aged 53 or younger.

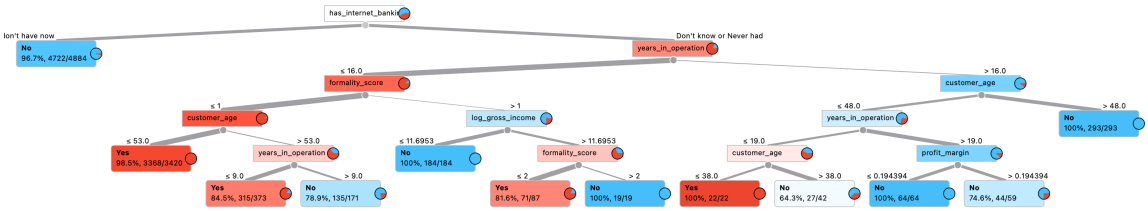

Figure 3: Binary Decision Tree for Large Informal Segment

The primary risk associated with this segment is compliance and regulatory exposure, these businesses possess the cash flow and operational history to survive, but they lack formality and have digital banking blindness that prevents the use of standard credit assessment models. We propose the below:

1.Digital Onboarding: Since lack of internet banking is the root driver for this segment, the most impactful risk mitigation is educational. The MSME's need to be onboarded into basic mobile wallet and digital banking platforms. By doing so, this transitions their cash transactions into visible and verifiable data systems.

2.Incentivized Formalization: Rather than rejecting these businesses for low formality scores, the lender should create some flexible credit products. For instance, the MSME's can be granted an initial working capital loan based on their cash-flow appraisals, with contract agreement that interest rates will dynamically reduce upon the business achieving some formality by providing legitimate registration and tax compliance documents.

1.5. Stable Formal Segment

This segment represents the most bankable and structurally sound demographic within the MSME demographic. The defining feature for the segment is operational registration. The root node dictates that a formality_score of 1 or below results in a 99.3% exclusion from this segment. Formal financial integration is a strict prerequisite. Businesses that possess high formality, have been operating for 17 years or less and are located in either DRC or Ghana have a high probability of being considered in this segment. The stable formal segment is heavily influenced by the geographical location and operational tenure. For instance, in markets like Kenya and Uganda, the barrier to be considered in this segment is higher. Businesses are required to have higher formality, combined with strong revenue efficiency.

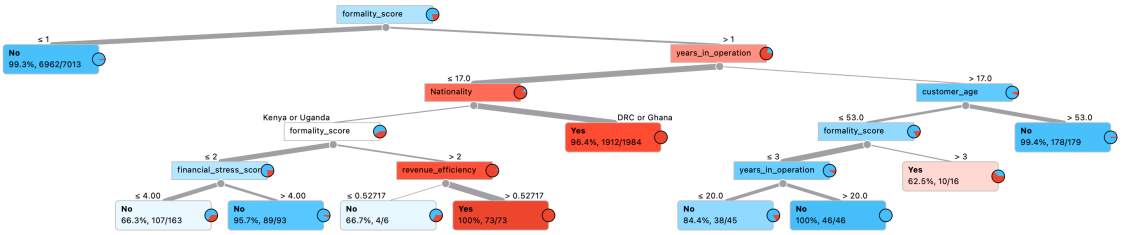

Figure 4: Binary Decision Tree for Stable Formal Segment

The primary risk associated with this segment is not default or compliance, it is customer churn. Since such businesses have verifiable financial histories and are highly stable, the lender must adopt an offensive retention strategy to compete with other financial institutions. We propose the below:

1.Seamless Customer Experience: The standard loan application procedure is too long and tedious for this tier. The lender should consider offering pre-approved loan processes, reducing the friction of borrowing via a one-click digital disbursement channel, that will prevent the MSME customers seeking faster capital from elsewhere.

2.Geography specific thresholds: Based on the regional split in this model, the algorithms cannot be one-size-fits-all.A business in the DRC or Ghana can be granted Stable Formal credit products when they meet standard formality requirements. However, applicants in Kenya and Uganda will require stricter revenue efficiency checks before unlocking premium credit tiers.